AURA BI: ENTERPRISE AI BUSINESS INTELLIGENCE PLATFORM

TECHNICAL SPECIFICATION & ARCHITECTURAL CASE STUDY

Author / Candidate:	Software Engineer Candidate (Interview Submission)
Target Evaluation:	Unada Labs Pvt Ltd - Technical Interview Assessment
Project Scope:	Multi-Agent Collaborative LLM Swarms, REST APIs, & Real-Time Client Dashboards
Document Version:	1.0.0 (Production Grade Release)
Compilation Date:	July 2026

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1. Project Introduction & Strategic Scope

AURA BI is a state-of-the-art Business Intelligence (BI) and AI consulting platform designed to automate the process of researching companies, evaluating market positioning, and recommending highly specific AI integration roadmaps. The system acts as a virtual consulting team, taking a company name as its sole input and delivering a complete, high-fidelity corporate report in seconds.

Core Objectives Achieved in This Project:

- **Automated Deep Research:** Eliminated the manual labor of scanning multiple web sources for basic corporate metadata, news, and executive bios.
- **Rigorous Strategic Analysis:** Standardized SWOT evaluations and peer comparisons to ensure objectivity and structure.
- **Consultant-Level Actionability:** Designed algorithms to propose ROI-driven, concrete AI opportunities addressing specific company bottlenecks rather than generic AI advice.
- **Multi-Format Exports:** Provided one-click compilation into print-ready PDF briefs and presentation-ready PowerPoint slides, created entirely in-memory for server efficiency.

2. System Architecture & Component Design

The system is designed using a decoupled, modern **Client-Server Architecture**. This separation ensures that the frontend presentation layer can scale independently of the high-latency AI graph agent nodes.

A. Frontend Architecture (React 19 / Vite / TypeScript):

The user interface was built to deliver a premium, responsive experience utilizing several advanced frameworks:

- **Tailwind CSS:** Used to establish a clean, dark-themed, glassmorphic UI design system. Custom layout classes provide responsive styling that adapts seamlessly to desktop and tablet viewports.
- **Framer Motion:** Utilized to build step-by-step entrance transitions, interactive card hover rotation effects, and a dynamic loading stepper that gives immediate feedback as each agent executes.
- **Recharts:** An SVG-based responsive charting library used to render dynamic financial forecasts, ROI models, and cost breakdowns directly on the client side.

B. Backend API Layer (FastAPI / Uvicorn):

FastAPI was selected as the backend service layer for its exceptional performance, native asynchronous support, and robust request validation using Pydantic schemas. It exposes endpoints for triggering searches, reading history, and fetching binary exports.

3. LangGraph Multi-Agent Swarm Design

The core analytical processing is managed by a sequential **LangGraph State Graph** consisting of six independent, specialized agents. LangGraph was chosen over simple sequential chain calls because it maintains a structured, thread-safe global state and allows nodes to contribute iteratively to a shared schema.

The 6-Agent Swarm Execution Workflow:

1. **Research Agent:** Gathers official corporate facts, industry sector tags, headquarters locations, geographic footprints, key executive names, revenue figures, and funding data via Tavily search.
2. **News Agent:** Conducts targeted searches to discover the 3 most significant recent updates, product launches, expansions, or mergers/acquisitions.
3. **Business Analyst Agent:** Consumes the compiled research and news to evaluate internal strengths, operational bottlenecks, and external market opportunities or threats, producing a SWOT matrix.
4. **Competitor Agent:** Compares the target company against top sector peers, extracting revenues, employee metrics, tech stacks, and market positionings (e.g. Leader, Niche).
5. **AI Consultant Agent:** Analyzes identified operational weaknesses and designs 3 custom, non-generic AI initiatives detailing problem, solution, implementation timeline, estimated cost, and ROI.
6. **CEO Pitch Agent:** Processes the entire accumulated state to generate a compelling executive summary headline, report synopsis, and a personalized pitch memo addressed to the target CEO.

Deterministic State Synchronization (Pydantic Mapping):

Each agent takes a shared, typed dictionary `AgentState` as input and returns modified fields. Upon graph completion, the accumulated state is structured directly into a strict `BIReport` Pydantic schema, guaranteeing that the frontend client receives fully structured, validated JSON data that conforms to TS interfaces.

4. API Integrations & External Data Pipeline

The platform acts as an orchestrator across multiple external web and LLM APIs to construct its analyses in real time:

- **Tavily Search API:** Integrated to perform search crawls. Tavily acts as a search agent, returning clean, markdown-formatted text summaries stripped of HTML boilerplate, which significantly reduces token consumption.
- **LangChain Core:** Standardizes prompt template mapping and handles connections to LLM providers. It dynamically loads Google Gemini (1.5 Flash) or OpenAI GPT models based on key availability.
- **ReportLab (PDF Export):** Compiles corporate data, SWOT grids, competitor tables, and CEO briefs into print-ready PDF attachments in-memory.
- **python-pptx (PowerPoint Export):** Programmatically constructs widescreen slide decks, automatically applying corporate theme layouts and dark-mode styles in-memory.

5. Engineering Challenges & Production Solutions

Building an autonomous multi-agent platform presented several technical challenges. Below is a breakdown of how they were resolved:

Identified Challenge	Implemented Solution
LLM Schema Deviation: LLMs frequently include conversational text (e.g. 'Sure, here is your JSON') or wrap payloads in markdown code blocks. This crashes standard JSON parsers.	Designed a custom wrapper function (*call_llm_json*) that enforces JSON-only constraints, cleans markdown backticks using regex, and parses payloads safely. If a failure occurs, the code falls back to pre-constructed models to prevent API failure.
Context Pollution: Search results for company names with multiple meanings (e.g. 'Apple' or 'Reliance') can pollute the model with mixed details, merging distinct subsidiary facts.	Injected strict context boundary instructions into the Research Agent prompt. The agent is forced to identify the primary well-known entity, ignore conflicting subsidiaries, and separate data categories like headquarters or revenue.
Cold-Start Experience: If a user runs the application without setting up API keys, empty dashboards or 500 crashes are returned, spoiling the initial review experience.	Built a hybrid fallback *Dynamic Demo Engine*. If credentials are missing, the system redirects to Demo Mode, serving high-fidelity presets for popular targets (Stripe, Tesla, NVIDIA, Apple, Jio) and generating deterministic mock templates for unknown terms.
Server Disk Bottlenecks: Writing PDF and PowerPoint files to the server's disk causes disk bloat, raises permissions issues, and slows down response times.	Routed all document compiler outputs into in-memory binary streams using Python's *io.BytesIO*. FastAPI streams these bytes directly as file downloads, ensuring zero disk overhead.

6. Summary & Key Highlights

This project successfully demonstrates the production-grade implementation of a multi-agent system, featuring robust error handling, seamless frontend-backend integration, dynamic charting, and multiple document export engines.

- **Developer Velocity:** Vite + TS ensures sub-second feedback during client development.
- **User-Centric Design:** Interactive dashboard features detailed SWOT analysis, competitor tables, and a scrollable letterhead CEO memo box.
- **Scalability:** Decoupled REST architecture allows integration with enterprise databases, search services, and LLM orchestration networks.